

1. Explain the following terms or results; Use specific example, formula, or graph if necessary.
 - (a) Explain the joint null and alternative hypotheses and give an empirical example.
 - (b) Explain the restricted and unrestricted regression models.
 - (c) Explain the formula and properties of the homoskedasticity-only F-test statistic.
 - (d) Show that for the OLS regression with multiple regressors, $R^2 = 0$ if $\hat{\beta}_1 = 0$, $\hat{\beta}_2 = 0$, ..., and $\hat{\beta}_k = 0$.
2. To test the effectiveness of a job training program on the subsequent wage of workers, we specify the model:

$$wage_i = \beta_0 + \beta_1 train_i + \beta_2 educ_i + \beta_3 experience_i + u_i, i = 1, \dots, n.$$

where $train_i$ is a binary regressor equal to 1 if the i -th person participated in the training program. Think of the error term as containing unobserved worker ability. Discuss in what kind of situation the OLS estimator of β_1 is likely to have bias and discuss the sign of the bias. In what kind of situation there will be no bias for the OLS estimator of β_1 ?

[Hint: This depends on how the employer assign the workers to the job training program (e.g., assigned randomly or not).]

3. Data were collected from a random sample of 200 home sales from a community in 2013. Let $Price$ denote the selling price (in \$1000s), BDR denote the number of bedrooms, $Bath$ denote the number of bathrooms, $Hsize$ denote the size of the house (in square feet), $Lsize$ denote the lot size (in square feet), Age denote the age of the house (in years), and $Poor$ denote a binary variable that is equal to 1 if the condition of the house is reported as “poor”. An estimated regression yields:

$$\widehat{Price} = 109.7 + 0.567BDR + 26.9Bath + 0.239Hsize + 0.005Lsize + 0.1Age - 56.9Poor.$$

(22.1) (1.23) (9.76) (0.021) (0.00072) (0.23) (12.23)

- (a) What is the predicted loss in value if a homeowner lets his house run down so that its condition becomes “poor”?
- (b) Suppose that a homeowner adds a new bathroom to her house, which increases the size of the house by 80 square feet. What is the predicted increase in the value of the house?
- (c) Is the coefficient on BDR statistically significantly different from zero at the 5% level? Explain. Construct the 95% confidence interval for the coefficient on BDR .

- (d) Suppose you would like to test the hypothesis that *BDR* and *Age* do not matter for the selling price. Write down the joint null and alternative hypotheses. Explain the restricted and unrestricted regressions in this case.

4. The following equation estimates the effect of computer usage on wages.

$$wage_i = \beta_0 + \beta_1 compwork_i + \beta_2 comphome_i + \beta_3 compwork_i \cdot comphome_i + u_i, \quad i = 1, \dots, n, \quad (1)$$

where u_i denotes the error term. The dummy variable *compwork* equals one if the individual uses a computer at work and zero otherwise. The dummy variable *comphome* equals one if the individual uses a computer at home and zero otherwise. Explain the meaning of $\beta_0, \beta_1, \beta_2$, and β_3 . [Hint: compare all the four different combinations of the values of the dummy variables *compwork* and *comphome*.]

5. Empirical Exercise: Use the dataset **CPS08.dta** uploaded in NTULearn.

- (a) Run a regression of average hourly earnings (*ahe*) on age (*age*). What is the estimated effect of age on earnings?
- (b) Run a regression of *ahe* on *age*, gender (*female*, equal to 1 if female, 0 otherwise), and education (*bachelor*, equal to 1 if university degree, 0 otherwise). Suppose that β_1 is the effect of age on earnings (after controlling for gender and education). Can we reject $H_0 : \beta_1 = 0$ at the 5% significance level? Explain your conclusion using the values of *t*-test, *p*-value, and confidence interval reported in the software.
- (c) Bob is a 26-year-old male worker with a high school diploma. Predict Bob's earnings using the estimated regression in (b). Jane is a 30-year-old female worker with a university degree. Predict Jane's earnings using this regression.